Finding The Time Constant of a Circuit

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Introduction

The purpose of this lab is to find the time constant,τ, in a RC circuit.This was done by connecting a function generator to a capacitor, then to a resistor, which was the connected to an osciloscope. We then adjusted the osciloscope to dsiplay a square wave. We then adjusted the vertical and horizontal nobs to align the graph displayed on the osciloscope.The y axis of the graph represents the voltage and the x-axis represents the time in seconds.From the graph displayed on the osciloscope, we gathered 10 data points for voltage and time. The final result of the experiment is as follows: τ=$1.4163 \pm 0.11729s$

The fundemental equation for an discharging RC circuit was used for this lab:

$$\tau = RC \tag{1}$$

In this equation, the time constant τ, is equal to the resistance R, times the capacitance C.The above equation can then derive the discharging capacitor equation below:

$$V = V_0 e^{\frac{-t}{RC}} \tag{2}$$

In this equation, the voltage V, is equal to the initial voltage V_0, times Euler's number e, to the power of time $-t$, divided by the resistor R, times the capicitor C.

$$S = \frac{1}{-\tau} \tag{3}$$

Equation (2), is in the form of the equation of a line in the form of $slope = \frac{1}{-\tau}$.

Methods

We started by connecting the function generator to a capacitor. The capacitor then connected to a resistor. Finally the resistor was connected to an osciloscope. We set the function generator to a square wave.The osciloscope then displayed a wave on an axis, The axis could be adjusted on the osciloscope using knobs which would adjust the graph horzontally and vertically. After we adjusted the graph appropriately we defined the axis as a class. The x-axis would represent time and the y-axis would represent voltage. We then wrote the down the values of on the axis for voltage and time.This process was repeated until we 10 data points for both V and T.

The value of $slope$ is determined from a least squares fit to time T plotted as a natural log function of voltage V. The value of τ is calculated using V and T. The uncertainty in τ is calculated by propagating the uncertainty for S in the calculation for τ. Unless otherwise noted, all uncertainties are reported using t-statistics for 95% confidence intervals and confidence bands. All calculations are performed in MATLAB[1].

[1] The Mathworks, http://www.mathworks.com

Data

The data is shown in Table 1. The voltage and time for this lab was measured using the osciloscope. The uncertainty of the function generator is $\pm$ .5 μs ,the uncertainty of the osciloscope would be $\pm$ .05 volts.

Table 1: Data

I $(amps)$	r (cm)
1.1	4
1.2	4.15
1.3	4.5
1.4	4.6
1.6	4.85
1.7	5
1.8	5.15
1.9	5.5
2.0	5.75

Results

The results for the fitted model are shown in Figure 1 The slope of the line and the value of the time constant is τ=$1.4163 \pm 0.11729s$

The results for the fitted model are shown are shown in the image below. The results show that our experiement is not comparable to the value of the RC circuit. If we input the values for the resistor($10\ k\Omega$) and the capacitor($68\ nF$) into the equation for the time constant $\tau = RC$, we get the value of $6.8e-4$. which is much less than what we calculated. This experment shows that the real world value may be different than the values on paper due to other factors we are not aware of or did not account for. My hunch is that it has something to do with the impedence of osciloscope.

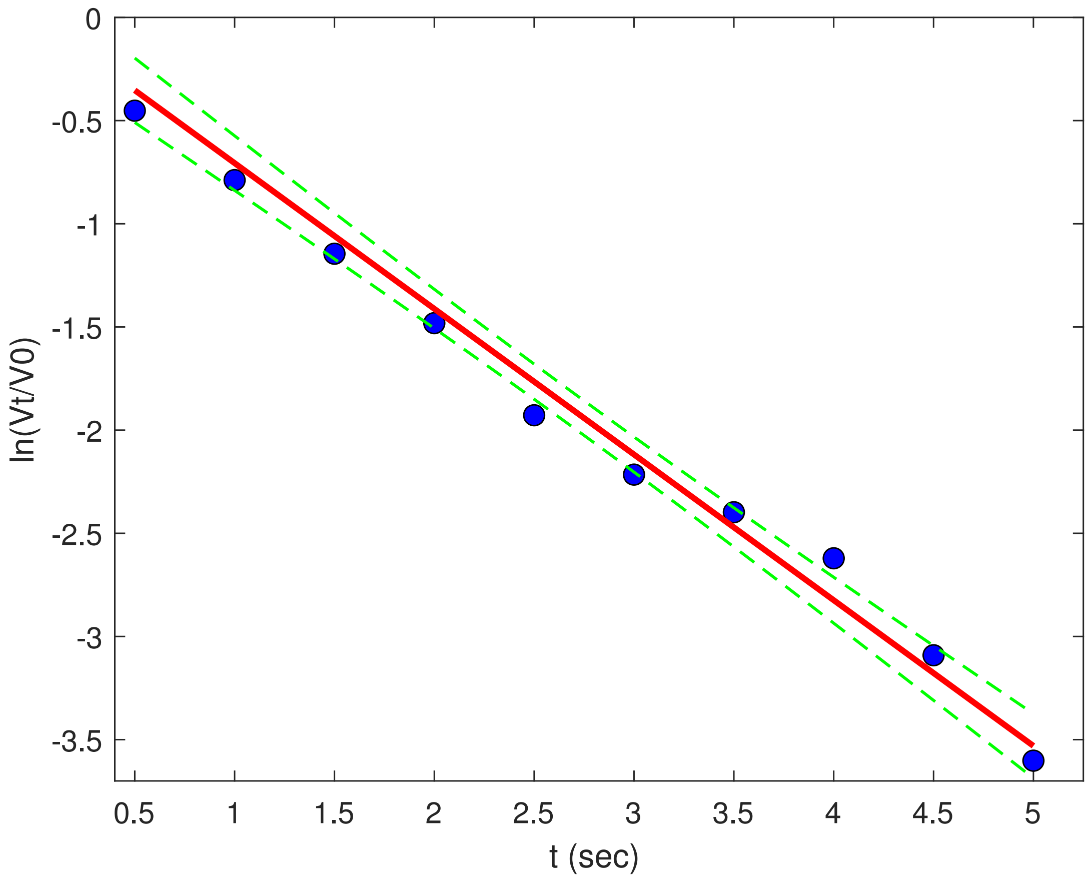

Figure 1: Trend line fit to $I(r)$. The data are shown as blue points. The fitted model is shown as a red solid line. The 95% confidence bands are shown as green dashed lines.